

4. Memory as Liability

1 point

A personal AI assistant remembers a user's preferences across weeks and uses them to make suggestions. Some preferences change over time. What is the primary risk introduced by long-term memory?

- ☐ Reduced creativity
- ☐ Increased latency
- ☒ Persistence of outdated or incorrect information
- ☐ Lower reasoning ability

5. Tool Overuse Detection

1 point

A team adds retrieval to reduce hallucinations but notices the AI now confidently repeats incorrect information from a flawed data source. What does this outcome most strongly suggest?

- ☐ Retrieval always improves accuracy
- ☐ The model's reasoning is broken
- ☒ Tools amplify both good and bad inputs
- ☐ Temperature should be lowered

6. No Tool Is the Right Tool

1 point

You ask an AI to brainstorm names for a new internal project with no factual constraints. Which approach is most appropriate?

- ☐ Retrieval from a document database
- ☐ Calculator-based evaluation
- ☒ Model-only generation
- ☐ Long-term memory persistence

☐ Long-term memory persistence

7. Compositional Tool Use

1 point

An AI system must answer factual questions, perform calculations, and explain its reasoning clearly to users. Which configuration best supports this task?

- ☐ Model-only reasoning
- ☐ Retrieval only
- ☒ Model reasoning + retrieval + calculator
- ☐ Fine-tuning without tools

8. Tradeoff After Improvement

1 point

A team adds retrieval to improve factual accuracy. Accuracy improves significantly, but response time doubles and API costs rise sharply. What is the most responsible next step?

- ☐ Remove retrieval immediately
- ☐ Ignore the latency because accuracy improved
- ☒ Evaluate routing or selective retrieval to balance accuracy and cost
- ☐ Increase temperature to compensate for speed

 You completed the Integrity Check for this submission. [View Integrity Check](#)

Your grade: 100%

Next item →

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1. When Determinism Matters

1 / 1 point

You ask an AI system to compute the total cost of an order with discounts, taxes, and currency conversion. Accuracy is critical. Which approach is mostappropriate?

✔ Nice work
2. When Tools Add Unnecessary Complexity

1 / 1 point

You ask an AI to summarize the themes of a short document you provide directly in the prompt. Which tool use is least appropriate?

✔ Nice work
3. Retrieval vs Reasoning

1 / 1 point

An AI assistant is asked: “What were the major outcomes of a policy change announced last month?” Which capability is most critical for a reliable answer?

✔ Nice work
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